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Hariom Bhargava, Mrityunjay Singh Chauhan & Yogendra Singh

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Food safety risks from soil-borne microplastics and antibiotic resistance across vegetable production and consumption pathways

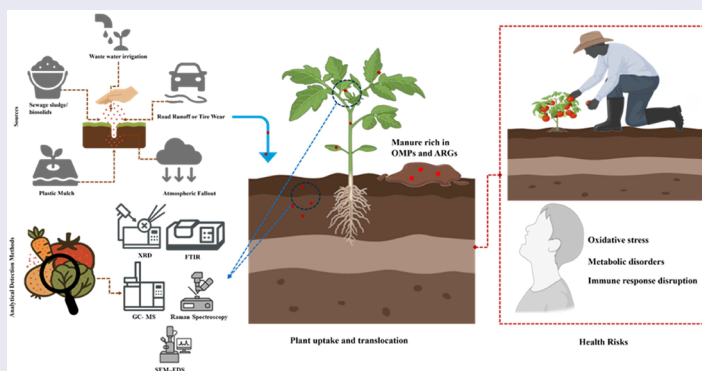
Hariom Bhargava, Mrityunjay Singh Chauhan, and Yogendra Singh

Department of Civil Engineering, Maulana Azad National Institute of Technology, Bhopal, India

ABSTRACT

Microplastics (MPs) increasingly infiltrate agricultural systems through plastic mulch degradation, wastewater irrigation, atmospheric fallout, and organic amendments. Once in soil, MPs interact with plant–soil interfaces: smaller particles adhere to root surfaces and translocate through apoplastic and symplastic pathways, while oversized microplastics (OMPs), in organic fertilizers represent a disproportionate fraction of total MPs mass and have been shown to impair crop growth by disrupting rhizosphere function and nutrient allocation. Physically alter soil porosity and root development. MPs also function as vectors for antibiotic resistance genes (ARGs) such as *blaTEM*, which adhere to polymer surfaces, migrate into plant tissues, and persist during digestion, raising concerns for food safety. However, inconsistencies in micro-particle isolation and characterization hinder process-based risk assessment. This review emphasizes the mechanistic pathways of MP entry, uptake, and biological interaction in agroecosystems, and highlights the urgent need for standardized detection protocols and food-grade thresholds to protect human health.

GRAPHICAL ABSTRACT



NOVELTY STATEMENT

This review presents a novel synthesis of microplastic contamination in agriculture by introducing oversized microplastics (OMPs, 1–5 mm) as an under-recognized but critical risk vector in organic fertilizers. It uniquely links OMP presence to measurable crop stress, systemic MP uptake, and the translocation of antibiotic resistance genes (ARGs) into edible plant parts.

Introduction

Microplastic (MP) pollution, once considered as a marine issue, is now recognized as a pervasive threat to terrestrial ecosystems, particularly agricultural soils and food chains (de Souza Machado *et al.* 2018; Li *et al.* 2025). Farmlands have become unintentional sinks for synthetic polymers due to plastic mulch degradation, wastewater irrigation, compost and biosolid application, and atmospheric deposition (Liu *et al.* 2023; Wang *et al.* 2023). These persistent particles are not passive contaminants; they interact with soil, water, and biota, altering physical, chemical, and biological processes.

Critically, MPs penetrate root systems, translocate through vascular tissues, and accumulate in edible plant parts, directly linking soil contamination to human dietary exposure (Myat *et al.* 2024; Ahmad *et al.* 2025).

Recent studies report MPs accumulation from roadside farms, with polyethylene (PE) and polypropylene (PP) being dominant. MPs carry additives, heavy metals, and persistent organic pollutants (POPs), which disrupt nutrient uptake, photosynthesis, and crop biomass production (Lalrinfela *et al.* 2024; Bolea *et al.* 2025). These stressors collectively reduce plant growth and compromise agroecosystem productivity.

Research interest has grown exponentially, with publications on agricultural MPs increasing from fewer than 10 papers in 2010 to over 2,000 by 2024 (Figure 1), reflecting the urgency of understanding this threat. Previous studies demonstrated that sludge and compost-amended soils accumulate MPs that can penetrate plant roots and translocate through xylem and phloem (Dhole *et al.* 2024). Together, these findings challenge the long-held assumption that plant barriers exclude synthetic particles, demanding a reevaluation of food safety risk frameworks, this work underscores the urgent need for standardized detection methods, regulatory thresholds, and mitigation strategies to protect agroecosystem integrity and public health.

Pathways of microplastic entry into agricultural ecosystems

Agricultural systems are increasingly recognized as significant reservoirs for MPs accumulation (Bodor *et al.* 2024; Li *et al.* 2025). Synthetic polymer particles infiltrate soils, water systems, and atmospheric interfaces, posing multi-pathway risks to agroecosystem function and food security (Prajapati *et al.* 2024). Recent research has illustrated that MPs enter agricultural matrices through several key and often interacting pathways. Among the most prominent is plasticulture, including mulch films and other plastic farming components, all of which degrade over time and release MPs directly into soils (Ghosh *et al.* 2025; Senevirathne *et al.* 2025).

PE-based mulch films, widely used to enhance yield and soil temperature regulation, fragment under UV exposure and mechanical stress, releasing MPs that persist across cropping cycles (Beriot *et al.* 2023; Salama and Geyer 2023). Residual fragments ($0.7\text{--}7.4\text{ mg kg}^{-1}$ in soil) reduce microbial activity and nitrogen cycling (Corradini *et al.* 2019; Rillig *et al.* 2019). Even so-called biodegradable films degrade

incompletely (<60% in six months), leaving polymer residues (Serrano-Ruiz *et al.* 2023).

Organic soil amendments such as compost, digestate, and sewage sludge are major but under-regulated sources of MPs, with studies consistently reporting significant contamination especially in composts derived from mixed municipal waste (Scopetani *et al.* 2022). Recent analyses indicate that MPs in composts originate from packaging residues and low-efficiency sorting their abundance can reach $10^2\text{--}10^3$ particles kg^{-1} (Li *et al.* 2020). Such contamination arises from the inclusion of plastic-laden household waste, packaging residues, and mismanaged litter in the composting process, which often escapes detection and removal during mechanical sorting stages.

Sewage sludge from wastewater treatment plants is a major MPs source for agricultural soils. Because primary and secondary treatments poorly retain small particles, sludge often contains very high concentrations frequently over 5,000 particles per kilogram (Corradini *et al.* 2019; Li *et al.* 2020; Franco *et al.* 2023). When applied as fertilizer or soil conditioner, sewage sludge becomes a key pathway for transferring synthetic polymers into agricultural soils. The main polymers found were PE, PP, Polystyrene (PS), and Polyvinyl Chloride (PVC) originate from household waste, textile fibers, personal care microbeads, and degraded packaging materials (Corradini *et al.* 2019; Biba *et al.* 2023).

Atmospheric deposition represents an often-overlooked input of MPs to cropland. In metropolitan Shanghai, deposition reached around $3,300$ particles $\text{m}^{-2}\text{day}^{-1}$ (Jia *et al.* 2022), dominated by Polyethylene Terephthalate (PET) and Polyamide (PA) fibers. However, this rate likely overestimates rural exposure representative multi-region data remain scarce. Textile abrasion, tire and brake wear, construction, and urban dust emissions significantly add to airborne MPs, making atmospheric transport a major pathway for MP accumulation, especially in open-field agricultural systems with minimal barriers to deposition (Periyasamy and Tehrani-Bagha 2022; Beaurepaire *et al.* 2025). Recent global observations confirm that atmospheric fallout contributes substantially to terrestrial MP loads, with deposition rates in rural and urban zones differing by over an order of magnitude (Bodor *et al.* 2024; Li *et al.* 2025). Such evidence strengthens the recognition of airborne MPs as a significant, though spatially variable, input to agricultural soils. Figure 2 illustrates MP pathways in agricultural ecosystems. Previous studies confirm that airborne transport is a key component of the terrestrial MP cycle, with atmospheric deposition rates ranging from 9.1 particles $\text{m}^{-2}\text{day}^{-1}$ in forests to $575\text{--}1,008$ particles $\text{m}^{-2}\text{day}^{-1}$ in urban gardens (Amato-Lourenço *et al.* 2025; Weber and Bigalke 2025), and global atmospheric resuspension from bare soils estimated at 104 tons yr^{-1} (Evangelou *et al.* 2024). Agricultural soils show MP concentrations of $95.68\text{--}100.93$ pieces kg^{-1} , with atmospheric deposition as a significant input pathway (Mahreen *et al.* 2025). These findings emphasize the need to incorporate atmospheric inputs into risk assessments, regulations, and monitoring programs to safeguard soil health and food security.

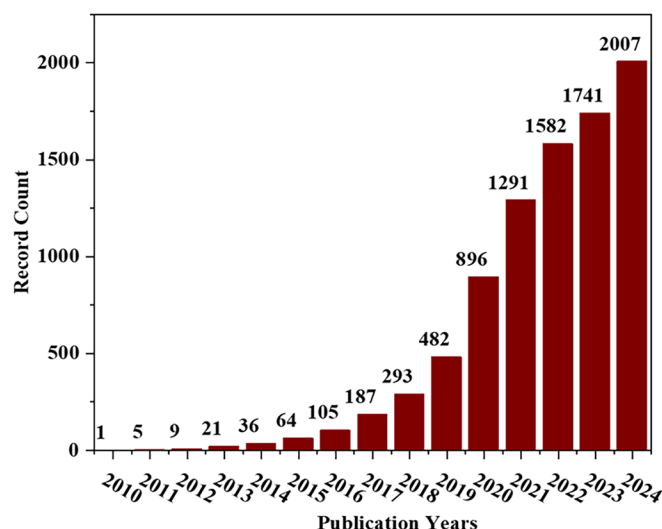


Figure 1. Annual publication output on microplastic research, based on a search in the Web of Science using the keyword ALL = (microplastic), refined for articles (2010–2024).



Figure 2. Visual representation of microplastic entry pathways in agricultural systems.

The combined inputs of mulch residues, biosolids, compost, and atmospheric fallout drive progressive MP accumulation in soils and crops. Without coordinated monitoring and regulatory measures, these inputs threaten soil health and food security (Colombini *et al.* 2022). Considering the impacts on soil health, crop yield, ecosystem function, and human health via the food chain, urgent coordinated action is needed. Priority should be given to developing and implementing high-resolution, spatially distributed monitoring tools to detect MPs at environmentally relevant levels (Aijaz *et al.* 2025). Policymakers must integrate MP risk assessments into regulations, promote sustainable plastic alternatives, and adopt circular economy approaches to reduce plastic at the source. Without urgent action, MPs could threaten food security and ecosystem health (Colombini *et al.* 2022; Mushtak *et al.* 2024).

MP characterization in environmental systems

The detection and characterization of MPs in environmental and food matrices rely on combined physical, chemical, and spectroscopic approaches to isolate and quantify polymers (Rathore *et al.* 2023). Yet inter-laboratory variability remains high, with identification errors exceeding 20% (Zhao *et al.* 2025). Complex matrices such as soils, composts, and food tissues require specific extraction approaches to avoid underestimation or false positives (Sharma *et al.* 2024). Common workflows integrate density separation, chemical or enzymatic digestion, flotation, and pressurized-fluid extraction

(PFE) (Rani *et al.* 2023; Tuuri *et al.* 2024). NaCl solutions recover 50–90% of low-density polymers, whereas ZnCl₂ achieves >95% efficiency in dense or fibrous matrices like banana tissue (Rajendran *et al.* 2022). Oxidative digestion with H₂O₂ or nitric acid provides 37–97% recovery in organic-rich samples (Corradini *et al.* 2019), and PFE recovers >90% of larger MPs from solid waste (Fuller and Gautam 2016). Rigorous contamination control, including procedural blanks, clean lab practices, and sample homogenization, enhances reliability (Todd *et al.* 2024; Seo *et al.* 2025).

Digestion methods aim to remove organics while preserving polymer integrity. H₂O₂ is preferred for its efficiency and polymer safety, sometimes assisted by surfactants (Pfeiffer and Fischer 2020). KOH may degrade sensitive polymers, and strong acids are generally unsuitable (Enders *et al.* 2017; Schrank *et al.* 2022). Enzymatic digestion is highly selective but slow and costly. Post-digestion, MPs are concentrated through density separation (NaCl, ZnCl₂) and membrane filtration (0.45–5 µm) (Singh *et al.* 2023; von Friesen *et al.* 2019). Spectroscopic tools micro-FTIR, ATR-FTIR, Raman, and SEM-EDS remain essential for polymer identification and morphological assessment (Rimal *et al.* 2024; Conterosito *et al.* 2025). However, recovery varies with matrix and particle size, limiting comparability (Matavos-Aramyan 2024; Ducoli *et al.* 2024). Standardized protocols and certified reference materials are urgently needed (Dey *et al.* 2025; Lin *et al.* 2025). Across agricultural matrices, PET, PS, PVC, and PE remain the dominant polymers detected (Gouda *et al.* 2025; Denaro *et al.* 2020). Table 1 summarizes extraction, recovery,

Table 1. Comparative overview of extraction, recovery, and characterization techniques for MP detection in agricultural and food matrices.

S. no.	Source	Methodology	Efficiency	Polymer size range (µm)	Matrix type	Recovery rate validation	Identification methods	Types of MPs detected	Environmental relevance	Reference
1	Soil	Oxidative digestion + density separation	37–97%	10–5,000	Organic	Yes	FTIR, Raman, SEM	PET, PS, PVC, PE	Real agricultural soil	Corradini et al. (2019) and Gupta et al. (2024)
2	Banana	Density separation using ZnCl ₂	>95%	–	Organic	Yes	FTIR, Raman	PE, PP, PET	Edible food product	Rajendran et al. (2022)
3	Soil compost	Oil-extraction protocol	>82%	10–5,000	Organic	Yes	FTIR, Raman, SEM	PET, PS, PVC, PE	Compost/fertilizer	Scopetani et al. (2022)
4	Brijjal	Density separation using NaCl	50–90%	–	Organic	Yes	FTIR, Raman	PE, PP	Edible crop	Pan et al. (2024)
5	Potato	Density separation using NaCl	50–90%	–	Organic	Yes	FTIR, Raman	PE, PP	Edible crop	Aydm et al. (2023)
6	Water	Filtration	98–99%	1–5,000	Inorganic	Yes	Micro-FTIR, Raman	PET, PP, PE	Drinking water	Jones et al. (2020) and Garfansa et al. (2025)
7	Soil	Plastic flotation + separator system	~40 µm	40–5,000	Organic	Yes	FTIR, Raman	PE, PP	Agricultural field soil	Mushtak et al. (2024)
8	Metalliferous waste	Magnetic separation (D2EHPA 0.08 M)	99%	–	Inorganic	Yes	FTIR, Raman	PE, PP	Industrial waste	Shi et al. (2022)
9	Air	Elutriation	93–98%	–	Inorganic	Yes	FTIR, Raman	PE, PP, PET	Ambient atmospheric air	Parga Martínez et al. (2023)
10	Solid waste	Pressurized fluid extraction (PFE)	>90%	500–5,000	Organic	Yes	FTIR	PVC, PE, PP	Mixed solid waste	Fuller and Gautam (2016)

and characterization techniques for MPs in agriculture and food matrices.

Mechanisms of MP uptake and translocation in crop plants

The entry and movement of MPs in plants link environmental exposure to food chain transfer. As shown in Figure 3, MPs penetrate plants via roots, stomata, and vascular tissues. The rhizosphere is the primary entry point, where MPs (100 nm to 10 µm) adhere to root surfaces and enter through high-permeability sites such as root hairs, lateral root zones, and damaged epidermis, as shown by confocal microscopy and fluorescent particle tracing (Li *et al.* 2023). In hydroponic setups, fluorescent PS beads penetrated cortical tissues within 48 h. Root mucilage and exudates enhance adhesion, while MP-induced microbial shifts weaken barrier integrity (Guo *et al.* 2022).

Once internalized, MPs move via apoplastic (up to 100 µm) or symplastic routes (endocytosis, ≤100 nm under stress). Experimental verification of apoplastic transport for particles ≥100 µm is limited; most evidence confirms movement of only nano- and sub-micron-sized plastics (Luo *et al.* 2022; Yu *et al.* 2024). Mobility increases for neutral, hydrophilic, spherical, or surface-oxidized particles. Vascular transport has been shown mainly in hydroponic and spiking systems. Nanoplastics (NPs) (<100 nm) are detected in xylem and phloem via micro-Raman and FTIR (Yu *et al.* 2024). Transpiration and osmotic flow facilitate upward movement, especially in seedlings. Uptake and translocation differ notably across plant types and developmental stages. Root crops such as carrot and onion exhibit MP retention mainly within the cortex, while leafy vegetables and seedlings allow greater systemic movement under stress. Water deficit and salinity increase MP permeation by 40–60%, reflecting stress-induced porosity changes (Dong *et al.* 2025; Yu *et al.* 2024). In tomato and lettuce, MPs appear in stems and leaves within 5–7 days (Serrano-Ruiz *et al.* 2023).

Uptake varies by species and stage. Root crops (onion, carrot) accumulate MPs near epidermis/cortex, while seedlings show higher permeability under stress. Water stress (drought/salinity) increases uptake by 40–60% (*e.g.*, spinach, radish). In potato, MPs alter nutrient levels (Na, Mg, Zn) and raise vitamin C, showing harm and adaptive responses (Kaiser *et al.* 2025). Zhang Y *et al.* (2025) and Zhang W *et al.* (2025) reported that terraced paddy fields' structure and water dynamics significantly affect MP migration, retention, and redistribution, highlighting topography's critical role in agroecosystems (Zhang Y *et al.* 2025). These findings raise concerns about food safety and human health, emphasizing the need for further study of MPs long-term effects on crops, plant physiology, and soil health.

MP mobility in soils is dynamic, driven by abiotic and biotic processes. Rainfall, irrigation, runoff, and erosion drive lateral and vertical transfer. PE MPs (53–500 µm) are ~20% exported via runoff, while the rest infiltrate up to 8 cm, mostly retained in the top 2 cm (Severe *et al.* 2025). Biotic processes, especially soil fauna, strongly mediate

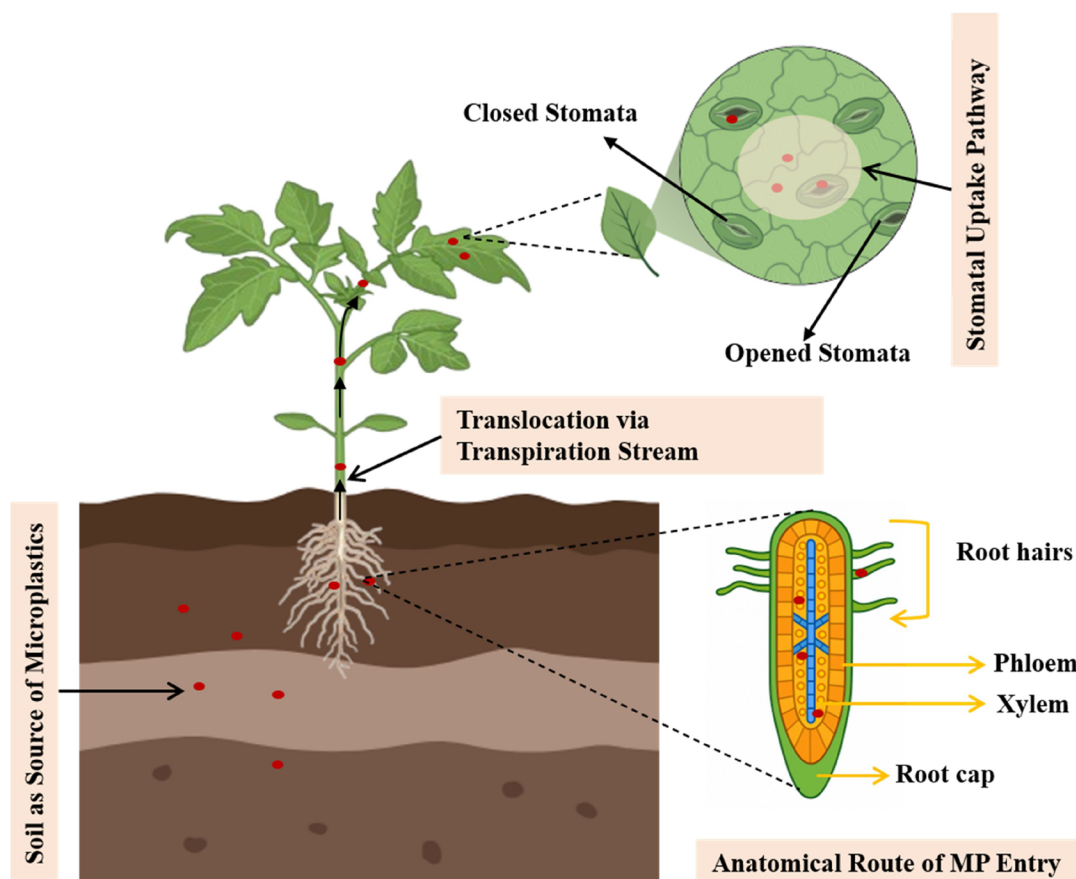


Figure 3. Microplastic entry and transport in plants via roots stomata and vascular tissues.

redistribution. Tracer experiments showed deep-burrowing earthworms ingest and excrete NPs (~256 nm), enhancing vertical MP motility and concentration along burrow walls (Heinze *et al.* 2021). Earthworm casting redistributes MPs, shreds biodegradable polymers like polybutylene adipate terephthalate (PBAT), and leaves inert plastics like low-density polyethylene (LDPE) intact (Adhikari *et al.* 2023). Overall, soils act as sinks and pathways for MPs, with physical processes depositing and mobilizing them, and biology relocating and reshaping them, affecting their long-term fate and ecological hazards. Quantitative thresholds for root penetration, vascular loading, and tissue retention remain undefined. High-throughput omics, micro-CT imaging, and isotopically labeled particles are needed to elucidate MP transport kinetics and plant response pathways (Maity *et al.* 2024; Zhang W *et al.* 2025). Future work should establish thresholds for root penetration and vascular loading to enable comparative risk assessment across crop types (Li *et al.* 2025).

Human exposure and health risks from MP ingestion

The ingestion of MPs through vegetables is now recognized as a major dietary exposure route worldwide, particularly in densely populated regions (Li *et al.* 2020; Canha *et al.* 2023). While seafood has historically been the focus of MP risk assessments, recent evidence indicates that vegetables may

contribute equal or greater exposure levels, given their broader consumption across populations. Figure 4 explains MP ingestion through raw vegetables and associated human health impacts.

The extent of dietary MP exposure through vegetables differs regionally due to irrigation sources, agricultural intensity, and consumption habits. Studies from Asia and Europe indicate that urban and peri-urban farms near wastewater inputs accumulate higher MP loads than rural zones (Sharma *et al.* 2023; Bodor *et al.* 2024). Across crops, leafy and root vegetables show greater accumulation than cereals or fruits because of direct soil contact and higher surface area (Rajendran *et al.* 2022; Aydın *et al.* 2023). This variation should be considered in the assessment of regional dietary MP exposure risks.

Exposure levels of MPs are influenced by environmental and agronomic factors. Vegetables grown near roads, in industrial zones, or irrigated with wastewater show higher MP loads. Compost and sludge application further elevate MP content, with studies finding up to 1,020 particles/kg in commercial compost (Scopetani *et al.* 2022; Aydın *et al.* 2023). Recent studies have also reported substantial MP contamination across a range of vegetables. Concentrations vary from 1,500 to 3,600 particles/kg in tomatoes, cucumbers, onions, and potatoes (Aydın *et al.* 2023) to markedly higher levels in lettuce (52,050), carrot (97,800), and broccoli (94,900) (Oliveri Conti *et al.* 2020). Spinach, cauliflower, and fenugreek contain 2,700–3,500 particles/kg (Ahmad *et al.*

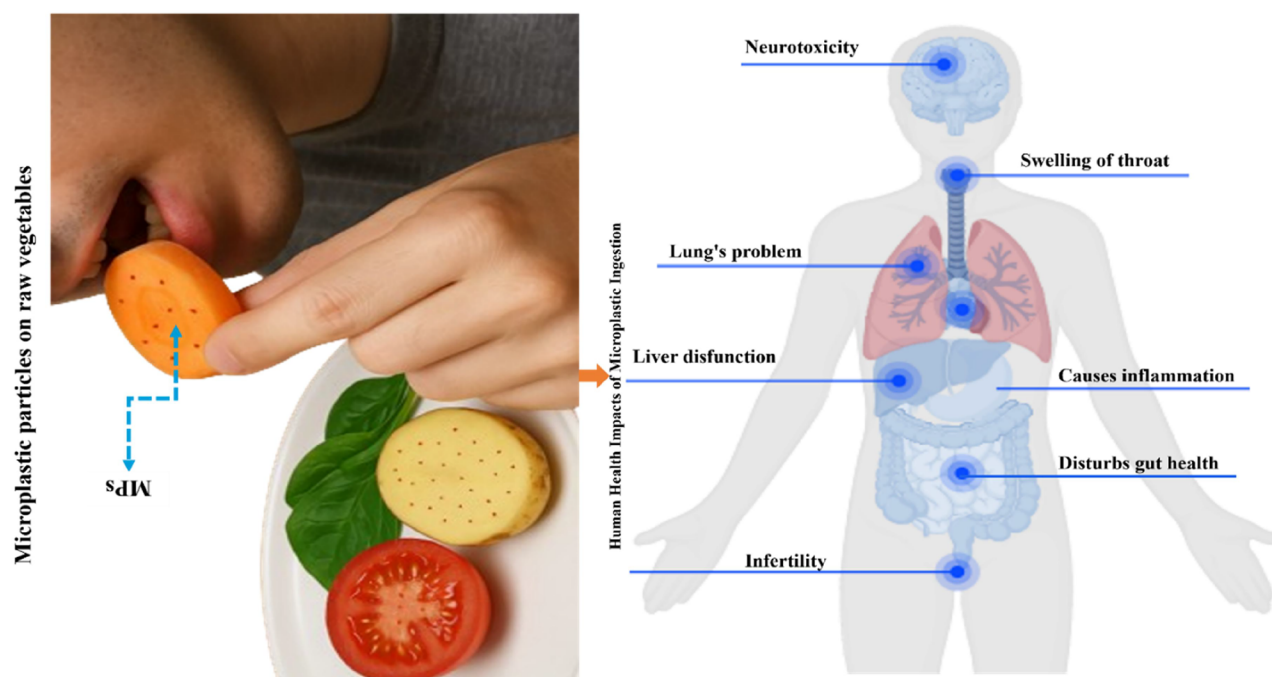


Figure 4. Microplastic ingestion through raw vegetables and associated human health impacts.

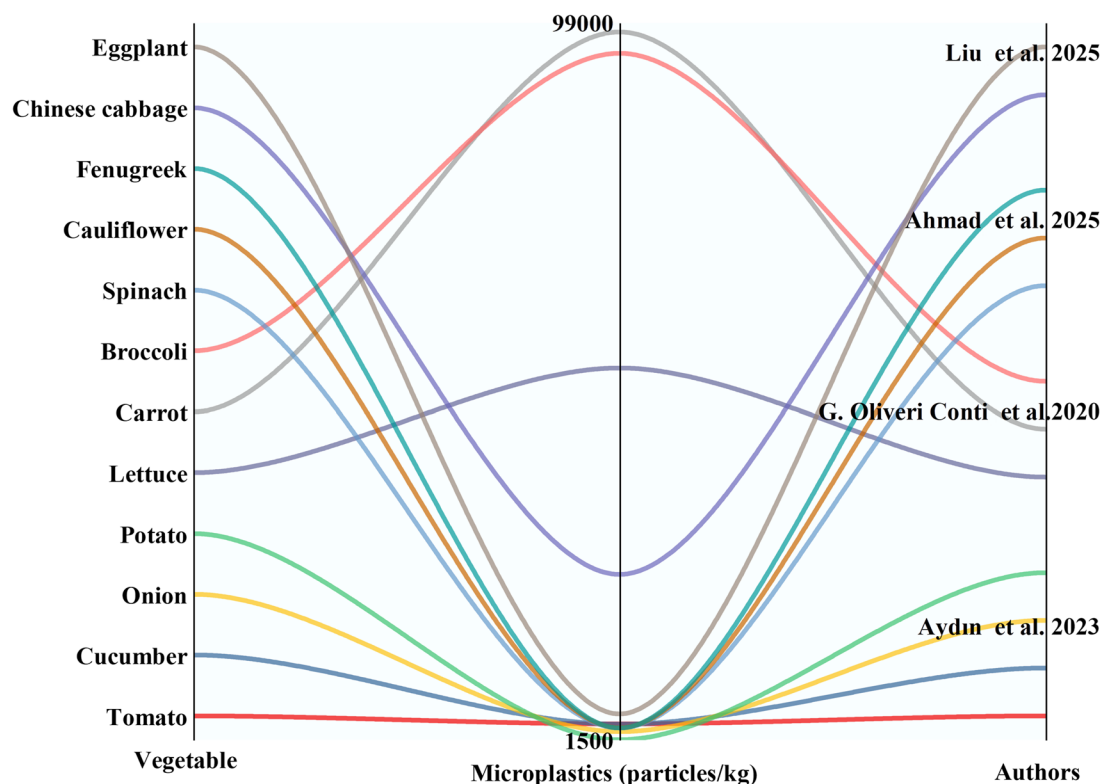


Figure 5. Based on previous studies, concentration of MPs in different vegetables.

2025), while Chinese cabbage and eggplant exhibit approximately 24,000 and 5,000 particles/kg (Liu *et al.* 2023), underscoring significant dietary exposure concerns (Figure 5).

MPs pose health risks by carrying additives (phthalates, bisphenols) and pollutants (heavy metals, polycyclic aromatic hydrocarbons (PAHs)). Experiments link them to inflammation, gut dysbiosis, and oxidative stress, while NPs may cross

epithelial barriers into systemic circulation (Giri *et al.* 2024; Lalrinfela *et al.* 2024). Dong *et al.* (2025) found significant associations between urinary MP concentrations and behavioral issues, including emotional problems, conduct disorders, hyperactivity, and peer difficulties (Dong *et al.* 2025). While the threshold dose for human toxicity remains undefined, chronic low-dose exposure raises concern, especially for

vulnerable populations like children and pregnant women (Halfar *et al.* 2023). You *et al.* (2025) showed PS-MPs reduce body weight, growth, and muscle quality in quails, and alter gut microbiota toward pro-inflammatory bacteria (You *et al.* 2025). MPs consistently impair gut and physical health across studies. Longitudinal research is needed to clarify causal links and dose–response effects.

Comparative analyses suggest vegetables may account for up to 35% of dietary MP intake in urban areas, surpassing bottled water and nearing seafood levels (Aydın *et al.* 2023; Chen *et al.* 2024). However, no regulatory framework currently defines safe MP limits in edible crops.

To mitigate risk, both upstream and downstream strategies are needed. These include reducing plastic use in agriculture such as phasing out non-degradable mulch, improving waste segregation and composting standards, and increasing consumer awareness on safe food handling (Li *et al.* 2014; Serrano-Ruiz *et al.* 2023). Additionally, standardized monitoring methods and longitudinal health studies are essential to translate exposure data into actionable guidelines.

Oversized microplastics in organic fertilizers: a hidden risk vector

Recent advances in MPs research highlight critical pathways through which oversized microplastics (OMPs) infiltrate agricultural systems and trigger sequential impacts on soil health, crop physiology, and food safety. Organic manure remains one of the most important MP entry routes, with concentrations reaching $129,800 \pm 82,300$ particles/kg in chicken manure; notably, OMPs constitute 75–100% of the total MP mass and represent the largest single contributor of

polymer bulk entering farmlands (Colombini *et al.* 2022; Chen *et al.* 2025). Due to their large size, structural rigidity, and low surface area-to-volume degradation kinetics, OMPs resist mechanical breakdown during composting, anaerobic digestion, and routine manure treatments (Colombini *et al.* 2022). As a result, they pass through fertilizer processing workflows unchanged, establishing themselves as long-lived particles in soil matrices. Their physical stability ensures prolonged contact with roots and microbial communities, making them a dominant and persistent MP fraction in agricultural soils (Ghosh *et al.* 2025).

Once introduced into soil, OMPs exert measurable effects on rhizosphere ecology. They alter root microbe signaling pathways, disrupt beneficial microbial consortia, and reduce plant height, root biomass, and harvest index in experimental systems, including lettuce (Chen *et al.* 2025). Root crops such as onions, carrots, and radishes are particularly vulnerable because OMPs remain localized in the upper soil layers where fine root networks proliferate. Their immobility within the soil makes them persistent interaction hubs that continuously influence nutrient cycling, water availability, and microbial community assembly. Figure 6 demonstrates the major contamination pathways through which MPs and OMPs move from manure into soil, crops, and eventually the food chain. While all MPs can act as plastisphere substrates, OMPs are especially concerning. Their large surface area and prolonged environmental persistence support the development of thick, multilayer biofilms characterized by dense microbial biomass and elevated loads of pathogens, antibiotics, and extracellular DNA. These stable plastispheres are hotspots for horizontal gene transfer (HGT), facilitating the movement of antibiotic resistance genes (ARGs) such as

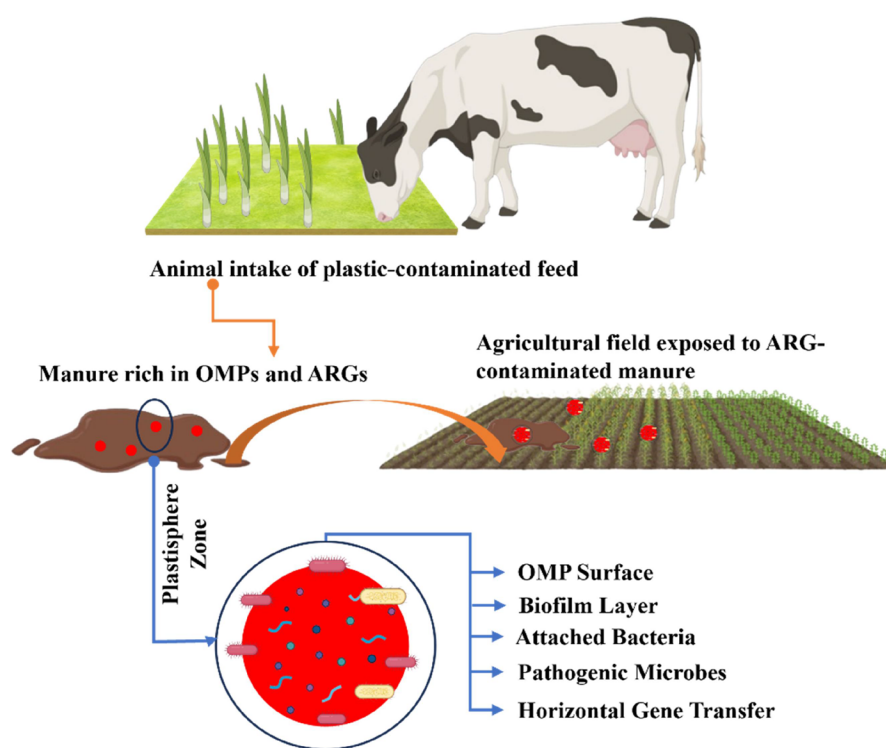


Figure 6. Pathway of microplastic contamination through the agricultural cycle.

blaTEM from manure-derived microbes to plant-associated bacterial communities. OMPs, therefore, function as high-capacity carriers that maintain ARG-rich microhabitats far longer than smaller MPs, amplifying their ecological and public health relevance (Rossi *et al.* 2025).

Current compost screening operations rely on mesh sizes of 10–20 mm, enabling the majority of OMPs to bypass filtration. Physical handling during composting or soil tillage may fragment some OMPs, producing smaller, more bioavailable MPs that disperse more readily through soil pores and root zones, further increasing ARG mobility. To prevent these outcomes, physical removal of OMPs must be paired with microbial inactivation and DNA-degradation methods such as thermophilic composting, alkaline stabilization, or oxidative disinfection (Xing *et al.* 2022; Colombini *et al.* 2024).

Plastisphere research consistently demonstrates that MPs including PE, PP, polylactide (PLA), and polycaprolactone (PCL) persist in soils and host distinct microbial assemblages enriched in pathogens and resistance determinants (Luo *et al.* 2025; Woodford *et al.* 2025; Wang *et al.* 2025). In vegetable fields, *Actinobacteria* and *Proteobacteria* dominate plastispheres at significantly higher proportions than bulk soils. Near wastewater treatment plants, viable *E. coli*, *Klebsiella spp.*, and *Citrobacter spp.* colonize MPs within 24 h, with biofilms remaining stable for up to 23 days (Bhardwaj *et al.* 2025; Rossi *et al.* 2025). Chemically, aged MP surfaces also act as adsorption platforms for environmental contaminants, retaining toxins such as okadaic acid at concentrations up to 8.0 pg/mm² (Pires *et al.* 2025). These biological and chemical loadings increase the likelihood of microbe-assisted MP uptake, as demonstrated by recent findings of MP accumulation in onion tissues. Although MPs pose clear agricultural hazards, their presence also provides valuable information about anthropogenic pressures, land-use patterns, and microbial stress responses. For instance, Chen *et al.* (2025) demonstrated that manure-derived OMPs modulate ARG expression and persistence even after simulated gastrointestinal digestion, illustrating their role as both contaminants and indicators. Importantly, their biological interactions not simply their abundance drive toxicity, including inflammation, dysbiosis, and systemic contaminant distribution in exposed organisms (Chen *et al.* 2025).

Given their durability, mass contribution, and ability to sustain ARG-rich biofilms, OMPs act as long-term reservoirs at manure–soil–crop–human interfaces. Their persistence amplifies One Health risks and underscores the need for OMP-specific monitoring, regulatory guidelines, and mitigation strategies within agricultural management frameworks.

Challenges in MP detection and management in agriculture

Despite advances in analytical methods and growing awareness, major challenges persist in detecting, measuring, and managing MPs in agriculture. A critical gap lies in identifying NPs (<100 nm), which pose greater biological risks due to their high mobility and ability to cross cell membranes. Common techniques like FTIR and Raman

spectroscopy lack the resolution and reference libraries needed for accurate detection of these tiny particles. Contamination during sampling and analysis remains a major issue, as airborne MPs and the use of plastic labware often lead to false results. Harsh chemical digestion methods can also degrade some polymers, causing underestimation of MP levels. The toxic effects of MPs are size-dependent, with smaller particles more likely to penetrate tissues, cause oxidative stress, and accumulate in organisms, yet they are addressed only to a limited extent in current studies. Additionally, there are no standardized global guidelines for MP testing or safe limits in crops, complicating data comparison and regulation.

In terms of management, plastic waste from agricultural practices like mulching film disposal often leads to environmental pollution, especially when farmers burn plastics due to cost or labor limits. Public awareness, particularly among smallholder farmers, remains very low, limiting efforts to reduce MP pollution or drive policy change.

Conclusions

Microplastics are emerging as major terrestrial pollutants threatening agricultural productivity, food safety, and public health. They enter agroecosystems via plastic mulch breakdown, biosolids, compost, wastewater irrigation, and atmospheric deposition. OMPs in organic fertilizers further amplify risks by impairing plant growth and promoting ARG transfer. These findings highlight an urgent need for standardized detection methods, clear safety thresholds, and sustainable agricultural practices to prevent long-term environmental and health impacts.

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Author contributions

Hariom Bhargava: Conceptualization, Investigation, Data curation, Methodology and Writing -original draft. Mrityunjay Singh Chauhan: Methodology, Resources, Conceptualization, Supervision and Writing - review & editing. Yogendra Singh: Investigation, Data curation, Software and Writing - review & editing.

Disclosure statement

The authors report there are no competing interests to declare.

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Data availability statement

Data will be made available on request.

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